



Chatbot Api Integration - Technical specification

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EurekaJO Chatbot API Integration – Technical Specification

1. Purpose of This Document

This document provides technical guidance for integrating a conversational AI chatbot with **eurekajo.com**. The chatbot will help users discover real estate listings using EurekaJO backend APIs, understand available options, and redirect users to search result pages on the website.

The intended audience is **backend / chatbot developers**.

2. High-Level Architecture

User → Chat UI → Chatbot Logic → EurekaJO APIs → Chatbot Response → User

The chatbot acts as an orchestration layer that: - Interprets user intent (buy/rent, category, location, filters) - Calls lookup APIs to validate values - Calls search APIs to retrieve results - Optionally generates a website URL for full browsing

3. Base Configuration

3.1 Base URL

<https://eurekajo.com>

3.2 Authentication

- All APIs listed below are **public** (no authentication required)
- Intended for **read/search** operations only

3.3 Response Format

- All APIs return **JSON**
 - UTF-8 encoded
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4. Lookup APIs (Static / Semi-Static Data)

These APIs should be cached by the chatbot to minimize requests.

4.1 Get Real Estate Categories

Endpoint

GET /api/LookUps/Categories

Purpose Returns all supported real estate categories.

Example Values - Apartment - Villa - Land - Farm - Office

Chatbot Usage - Map user input ("apartment", "flat", "شقة") to category ID -
Validate category before search

4.2 Get Offering Types

Endpoint

GET /api/LookUps/OfferingTypes

Purpose Returns transaction types.

Expected Values - Buy - Rent

Chatbot Usage - Detect intent: buying vs renting - Mandatory parameter for all searches

4.3 Get Predefined Areas

Endpoint

GET /api/LookUps/Areas

Purpose Returns predefined areas stored in EurekaJO system.

Important Notes - Each area includes a **polygon (geographic boundary)** -
Used for precise location-based searches

Chatbot Usage - Offer suggestions when user types known areas - Prefer this API over free-text search when match is found

4.4 Search Areas (Free Text Location Search)

Endpoint

GET /api/LookUps/SearchAreas

Purpose Searches locations using: - EurekaJO internal areas - Google Maps locations

Input - Area name (partial or full text)

Chatbot Usage - When user enters an unknown or informal location name -
Example: "near Mecca Mall", "Abdoun", "Downtown Amman"

5. Core Real Estate Search APIs

5.1 Search Real Estates

Endpoint

POST /api/Realestates/Search

Purpose Returns a list of real estate listings matching the search criteria.

Common Parameters (non-exhaustive) - CategoryId - OfferingTypeId (Buy / Rent) - AreaId or Geo Polygon - PriceFrom / PriceTo - AreaSizeFrom / AreaSizeTo - Bedrooms - Bathrooms - PageNumber / PageSize

Full parameter definitions are available in the Postman collection.

Chatbot Usage Flow 1. Extract structured filters from conversation 2. Validate values using lookup APIs 3. Call search endpoint 4. Summarize results (count, price range, top listings)

Recommended Chatbot Response - Total number of matching properties - 2–3 highlighted listings - Option to view all results on website

5.2 Generate Search Result Page URL

Endpoint

POST /api/Realestates/GetSearchURL

Purpose Generates a **public website URL** that displays the full search result set.

Inputs - Category - Offering Type - Area / Location - Search filters

Output - Fully constructed EurekaJO web URL

Chatbot Usage - Final step in conversation - Redirect user to browse listings visually

Example Chatbot Message > “I found 42 apartments for rent in Abdoun. You can view all results here: [URL]”

6. Suggested Conversational Search Flow

1. Greeting
2. Identify intent (Buy / Rent)
3. Identify property type
4. Identify location

5. Ask optional filters (budget, bedrooms)
 6. Call Search API
 7. Summarize results
 8. Generate & share website URL
-

7. Error Handling & Edge Cases

Scenario	Chatbot Behavior
No results	Suggest nearby areas or different budget
Invalid category	Offer valid categories
Unknown location	Use SearchAreas API
Missing required info	Ask clarifying question

8. Performance & Caching Recommendations

- Cache lookup APIs for **24 hours**
 - Cache area search results for **short-term (5–10 min)**
 - Avoid calling Search API until required parameters are confirmed
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9. Security & Rate Limiting

- Public APIs – do not expose internal admin APIs
 - Implement chatbot-side rate limiting
 - Log chatbot searches for analytics and improvement
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10. Deliverables for Chatbot Developer

- API base URL
 - This document
 - Postman collection
 - Sample request/response payloads
 - Expected conversational UX
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11. Future Enhancements (Optional)

- Save user preferences
- Personalized recommendations
- Multi-language support (Arabic / English)
- Lead capture & agent handoff